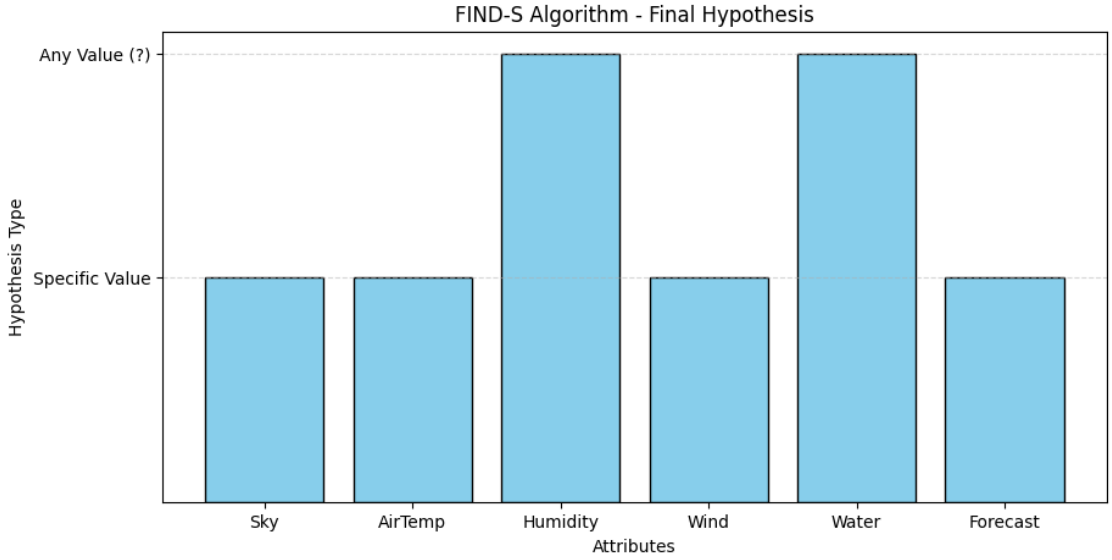
EXP : 1 OUTPUT :

```
Training Dataset:
     Sky AirTemp Humidity    Wind Water Forecast EnjoySport
0  Sunny    Warm   Normal  Strong  Warm     Same        Yes
1  Sunny    Warm     High  Strong  Warm     Same        Yes
2  Rainy    Cold     High  Strong  Warm   Change         No
3  Sunny    Warm     High  Strong  Cool     Same        Yes

Steps of FIND-S Algorithm:
After positive example 1: ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same']
After positive example 2: ['Sunny', 'Warm', '?', 'Strong', 'Warm', 'Same']
After positive example 4: ['Sunny', 'Warm', '?', 'Strong', '?', 'Same']

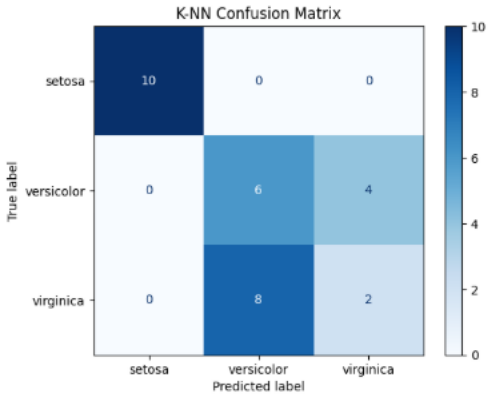
Final Most Specific Hypothesis:
['Sunny', 'Warm', '?', 'Strong', '?', 'Same']
```

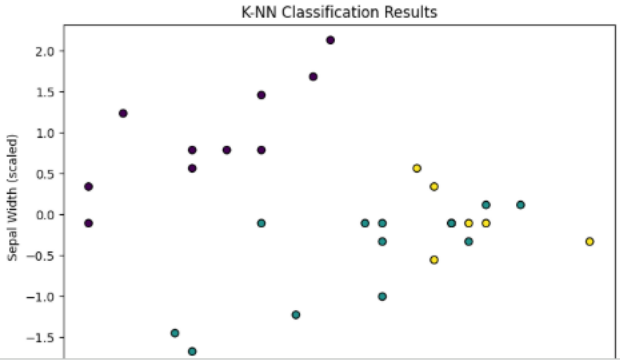


EXP : 2 OUTPUT :

```
K-Nearest Neighbours Algorithm
Number of training samples: 120
Number of testing samples: 30
Accuracy: 60.0 %

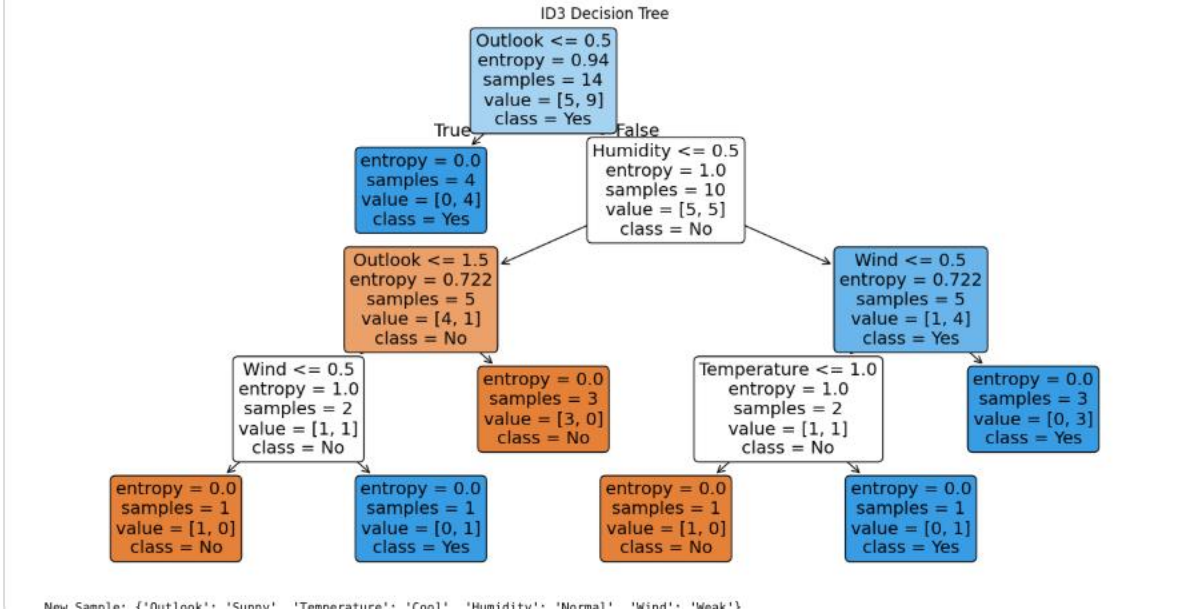
New Sample: [5.1, 3.5]
Predicted Class: setosa
```





EXP : 3 OUTPUT :

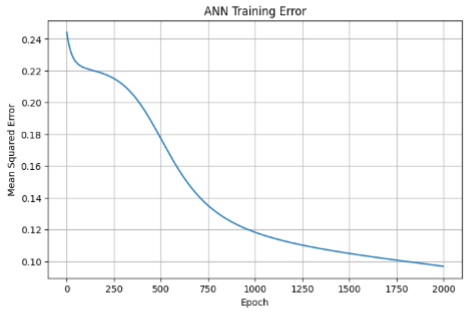
```
Training Dataset:
    Outlook Temperature Humidity    Wind Play
0     Sunny         Hot     High    Weak   No
1     Sunny         Hot     High  Strong   No
2  Overcast         Hot     High    Weak  Yes
3      Rain        Mild     High    Weak  Yes
4      Rain        Cool   Normal    Weak  Yes
5      Rain        Cool   Normal  Strong   No
6  Overcast        Cool   Normal  Strong  Yes
7     Sunny        Mild     High    Weak   No
8     Sunny        Cool   Normal    Weak  Yes
9      Rain        Mild   Normal    Weak  Yes
10    Sunny        Mild   Normal  Strong  Yes
11 Overcast        Mild     High  Strong  Yes
12 Overcast         Hot   Normal    Weak  Yes
13     Rain        Mild     High  Strong   No
```

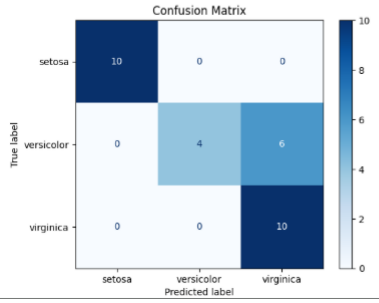



```
New Sample: {'Outlook': 'Sunny', 'Temperature': 'Cool', 'Humidity': 'Normal', 'Wind': 'Weak'}
```

EXP : 4 OUTPUT :

```
Artificial Neural Network using Backpropagation
Number of training samples: 120
Number of testing samples: 30
Test Accuracy: 80.0 %
```

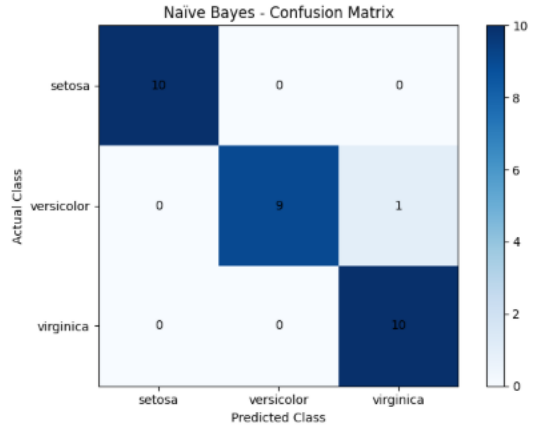




EXP : 5 OUTPUT :

```
Accuracy: 96.67 %

Confusion Matrix:
[[10  0  0]
 [ 0  9  1]
 [ 0  0 10]]
```




```
Classification Report:
              precision    recall  f1-score   support

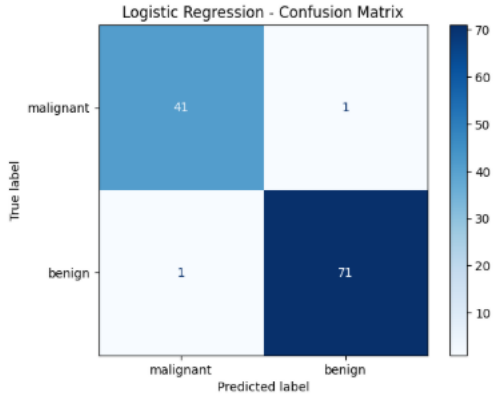
      setosa       1.00      1.00      1.00        10
  versicolor       1.00      0.90      0.95        10
   virginica       0.91      1.00      0.95        10

    accuracy                           0.97        30
   macro avg       0.97      0.97      0.97        30
weighted avg       0.97      0.97      0.97        30
```

EXP : 6 OUTPUT :

```
Accuracy: 98.25 %

Confusion Matrix:
[[41  1]
 [ 1 71]]
```




```
First 10 Predictions:
Predicted: [0 1 0 1 0 1 1 0 0 0]
Actual:    [0 1 0 1 0 1 1 0 0 0]
```